COMPSCI 572.01 (NLP)
MoEs + Prompting

Announcements

- Please submit your project team information by midnight tonight
  - Form link on Ed
  - Project details will be released later this week

- **Prof Monica Agrawal** will give Friday's lecture on post-training
  - I will be out of town

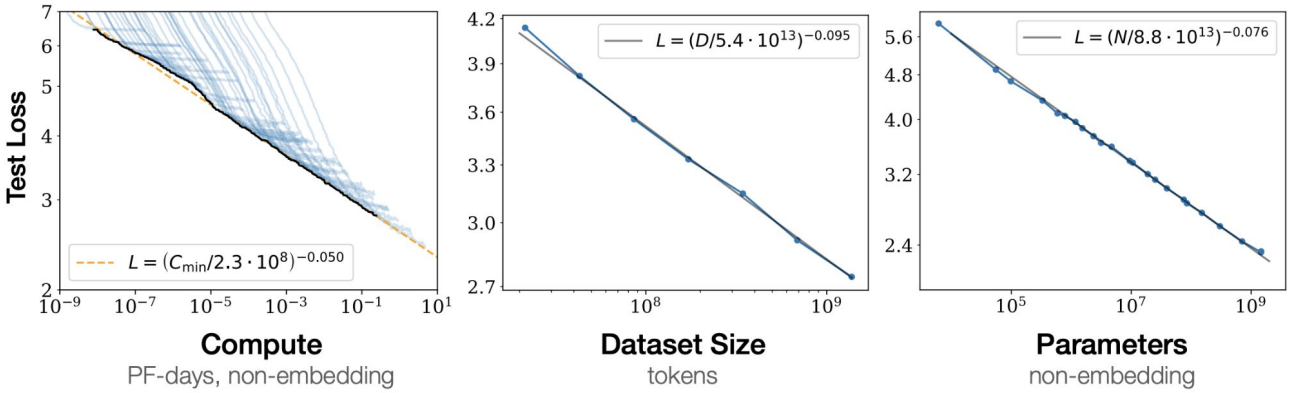
Overview

- Mixture-of-Expert (MoE) models
  - MoE Transformers
  - Routing functions and load balancing
- Prompting
  - In-context learning
  - Chain-of-thought
  - Self-consistency

Recap: Scaling laws

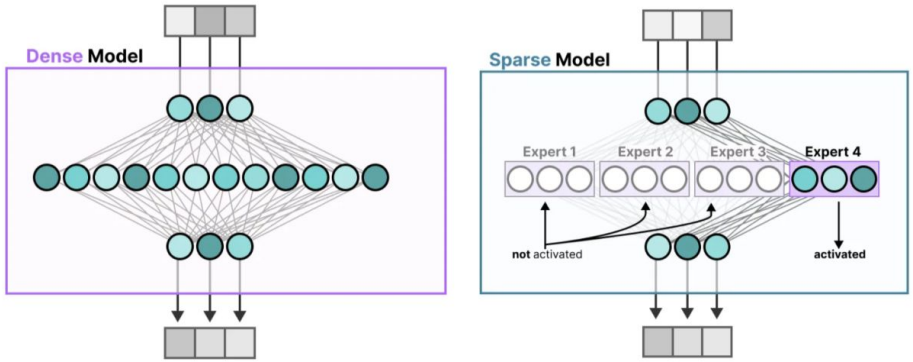
- Power law relationship between test loss and model size, training data size and training compute

- But for deployed LLMs inference compute is a bigger cost
  - So common to see smaller models trained for longer (as compared to what scaling laws suggest)



Idea behind mixture of experts

Conditional computation: Activate different parts of a neural network based on the input (Jacobs et al, 1991)

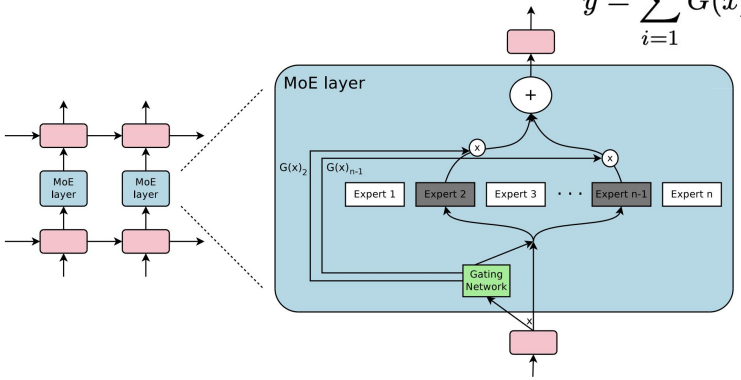


This allows scaling model size without increasing compute
(both training and inference)

Mixture-of-Experts (MoE) layer

Shazeer et al, 2017 – first application of conditional computation to scale NNs 1000x

$$y = \sum_{i=1}^{n} G(x)_i E_i(x)$$



Each expert is a FFN

The gating network selects the top k experts to **route** the input through

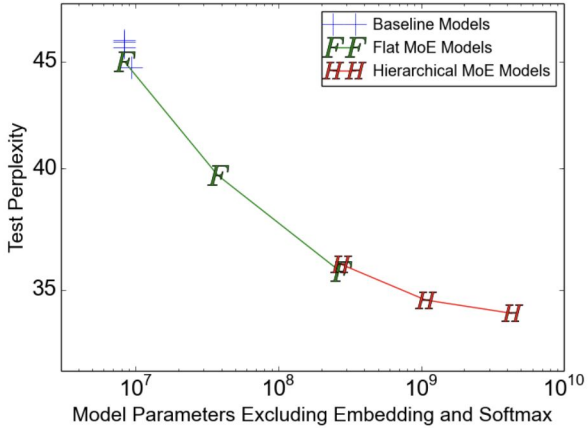
$$G_\sigma(x) = Softmax(x \cdot W_g)$$

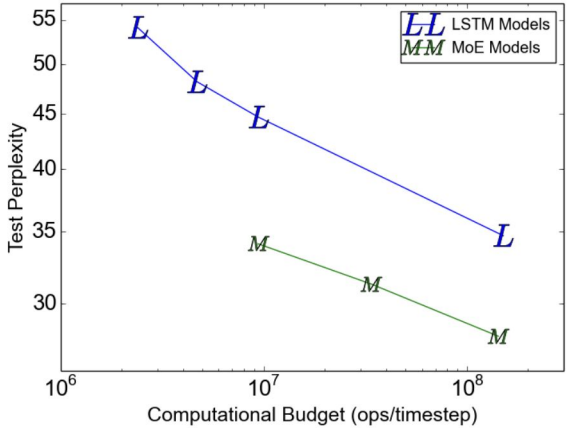
MoE layer applied between two LSTM layers (this was 2017)

Results (Shazeer et al, 2017)

State-of-the-art on 1B word LM dataset and competitive on machine translation datasets

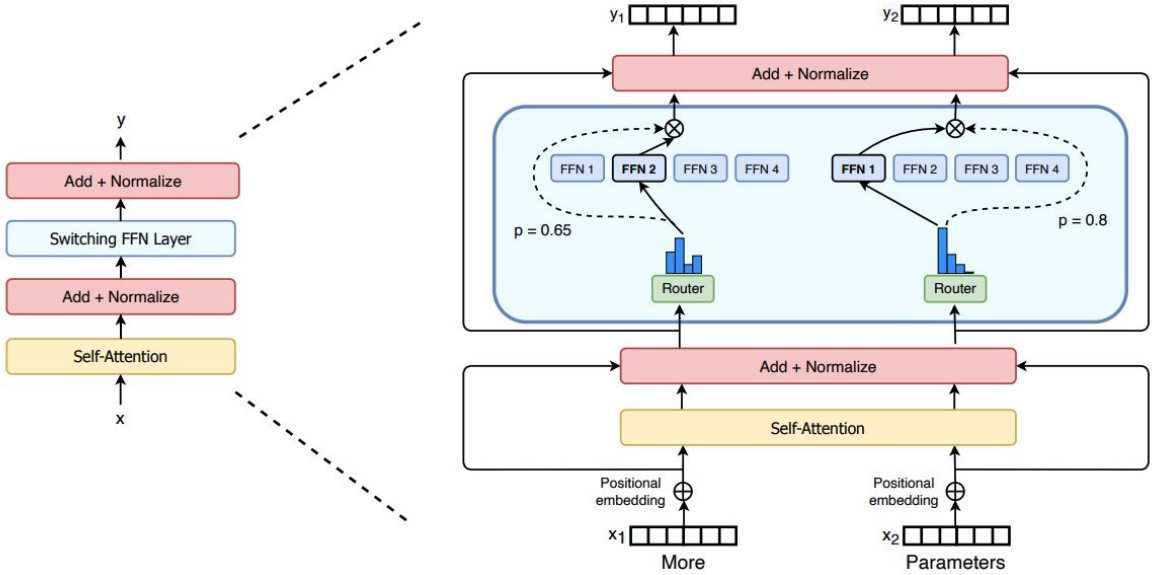
Beats LSTM models at the same computation budget (ops for one forward pass)





MoE Transformer

Replace the FFN in the transformer layer with a MoE



$$h(x) = W_r \cdot x$$

$$p_i(x) = \frac{e^{h(x)_i}}{\sum_j^N e^{h(x)_j}}.$$

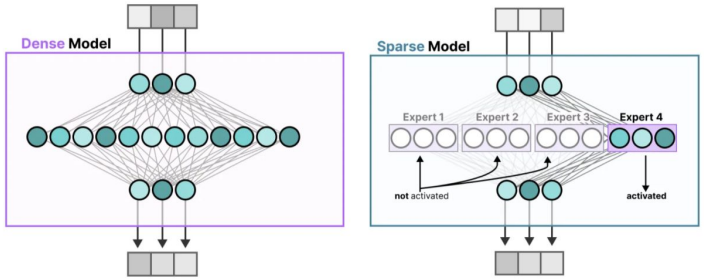
$$y = \sum_{i \in \mathcal{T}} p_i(x) E_i(x).$$

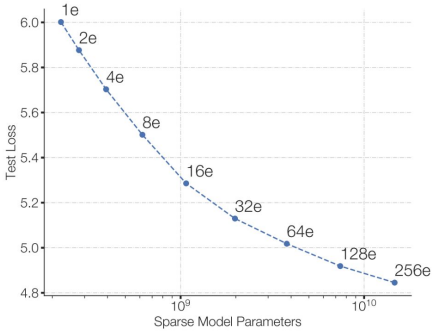
Switch Transformer, Fedus et al, 2022

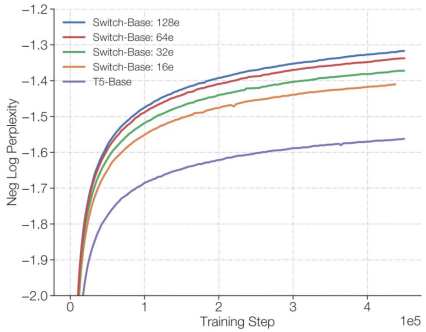
Most LLMs today are MoEs

Why?

- Increasing parameters improves test loss
  - Without adding compute







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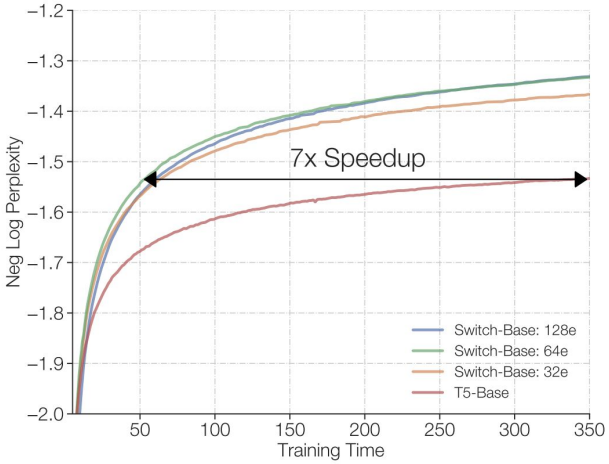
- MoEs are faster to train



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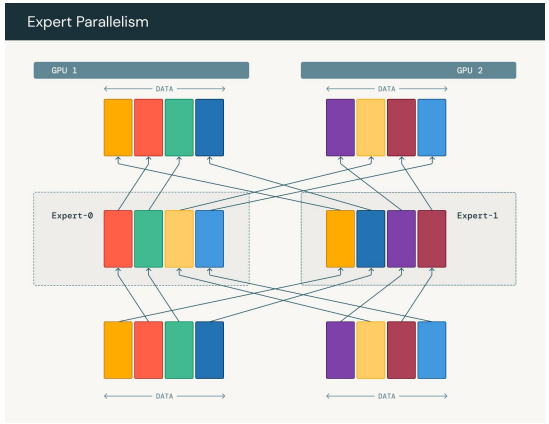
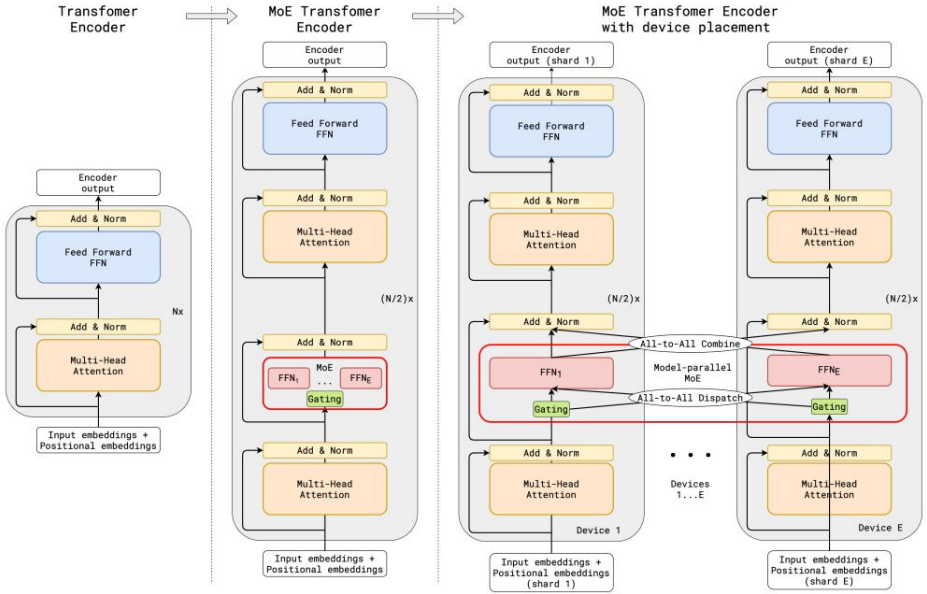
- Expert parallelization





GShard (2021)

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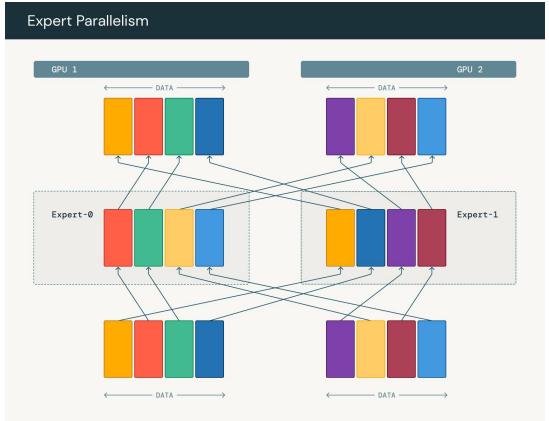

This can be more efficient than tensor parallelism – provided all experts are utilized properly.

GShard (2021)

Problem with training MoEs

- **Load balancing**: without special auxiliary losses, only a few experts are selected most of the time
  - This leads to poor utilization of GPUs

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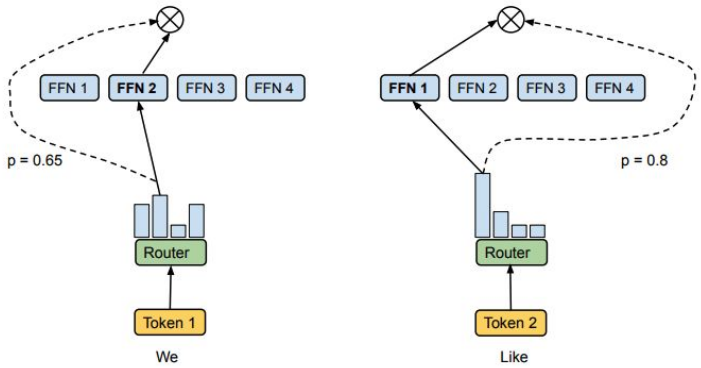
- **Solution:** Auxiliary losses which encourage balanced utilization of experts
  - **Load loss:** (Fedus et al, 2022) encourages uniform routing

$$\text{loss} = \alpha \cdot N \cdot \sum_{i=1}^{N} f_i \cdot P_i \qquad f_i = \frac{1}{T} \sum_{x \in \mathcal{B}} \mathbb{1}\{\text{argmax}\, p(x) = i\} \qquad P_i = \frac{1}{T} \sum_{x \in \mathcal{B}} p_i(x).$$

Gate values

Routing function

- Token-choice routing: each token selects its top-k experts

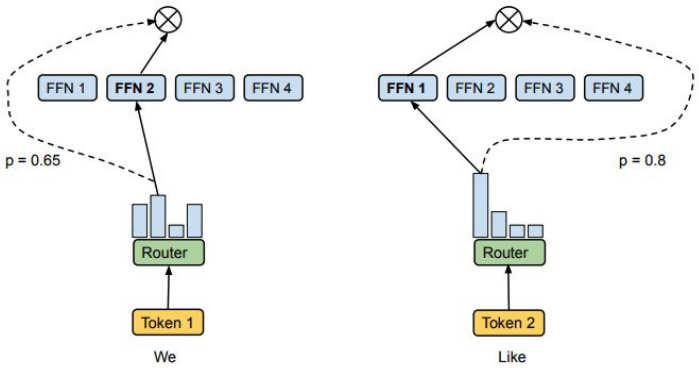


Token-choice

Routing function

- Token-choice routing: each token selects its top-k experts

- Problems:
  - Need to tune the aux loss
  - Still doesn't guarantee balancing, so a capacity factor is needed
  - Dropped tokens and wasted compute
  - Each token has the same number of experts



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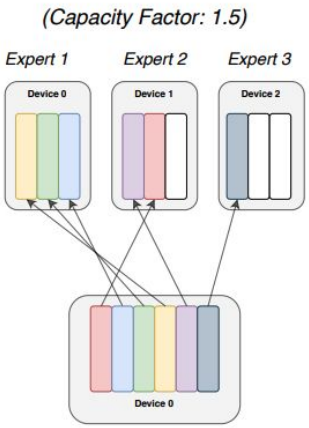
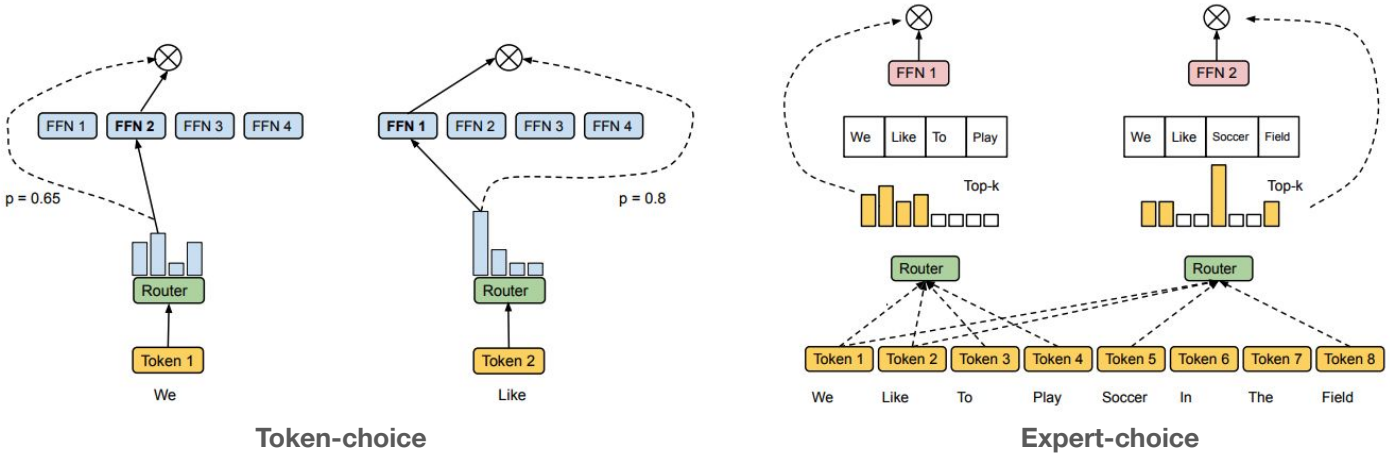


Routing function

- Expert-choice routing: each expert selects k tokens
  - Guarantees load balancing – no auxiliary loss needed
  - Allows variable number of experts per token
- But not good for generation – why?



Summary

- MoEs are sparse neural networks – each input activate a subset of parameters
- Due to this they are more efficient at inference – parameters can be increased without increasing FLOPs
- Routing functions are hard to optimize – but auxiliary loss functions help

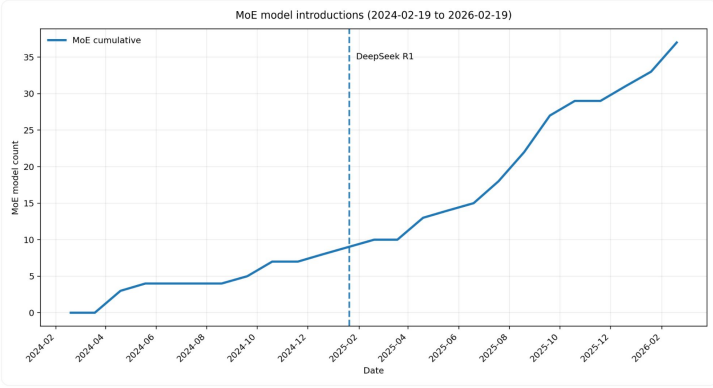


Figure 3: 2-year timeline of MoE model addition to the `transformers` library. DeepSeek R1 marks a clear inflection point.

https://huggingface.co/blog/moe-transformers

Prompting

GPT-2: zero-shot learning via prompting

- GPT-2: decoder-only model (1.5B parameters) trained on 40GB of internet text
- **Zero-shot learning**: frame text generation tasks as conditional language modeling problems and use model without fine-tuning

Question Answering	**Q:** Who wrote the book the origin of species? **A:**
Summarization	A magnitude 4.2 earthquake shook the … **TL;DR:**
Translation	**French:** Un homme expliquait que le fonctionnement … **English:**

- This is an example of *prompting*

Prompting recipe

- Given a language model $P(x;\theta)$
  - This need not be a proper LM – it could also be a masked LM $P(y|\tilde{x};\theta)$
- A prompting function $f_{\text{prompt}}(x', z)$
  - For example, in the case of sentiment: *[x'] The movie is [z].*
- And an answer space: $\mathcal{Z}$
  - For example, in the case of sentiment {“excellent”, “good”, “OK”, “bad”, “horrible”}
  - Also called “verbalizers”
  - This could be all possible natural language strings (generation)
- Search for: $\hat{z} = \text{search}_{z \in \mathcal{Z}} P(f_{\text{prompt}}(x', z); \theta)$

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  - This could be all possible natural language strings (generation)
- Search for: $\hat{z} = \text{search}_{z \in \mathcal{Z}} P(f_{\text{prompt}}(x', z);\theta)$ ⇦ Inference techniques we studied for seq2seq (greedy decoding, beam search etc)

How to construct the prompts?

- Manually

Task	Input ([X])	Template	Answer ([Z])
Sentiment	I love this movie.	[X] The movie is [Z].	great fantastic ...
Topics	He prompted the LM.	[X] The text is about [Z].	sports science ...
Intention	What is taxi fare to Denver?	[X] The question is about [Z].	quantity city ...
Aspect Sentiment	Poor service but good food.	[X] What about service? [Z].	Bad Terrible ...
NLI	[X1]: An old man with ... [X2]: A man walks ...	[X1]? [Z], [X2]	Yes No ...
NER	[X1]: Mike went to Paris. [X2]: Paris	[X1] [X2] is a [Z] entity.	organization location ...
Summarization	Las Vegas police ...	[X] TL;DR: [Z]	The victim ... A woman ... ...
Translation	Je vous aime.	French: [X] English: [Z]	I love you. I fancy you. ...

How to construct the prompts?

- Automatically, via gradient descent
  - Assuming we have some training data $\{x^{(i)}, y^{(i)}\}_{i=1}^{N}$

Original Input $\boldsymbol{x}_{\text{inp}}$

a real joy.

Trigger Tokens $\boldsymbol{x}_{\text{trig}}$

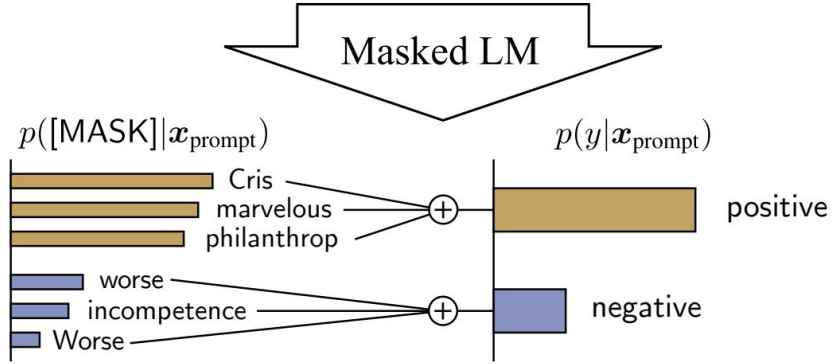
atmosphere, alot, dialogue, Clone...

Template $\lambda(\boldsymbol{x}_{\text{inp}}, \boldsymbol{x}_{\text{trig}})$

{sentence}[T][T][T][T][T][P].

AUTOPROMPT $\boldsymbol{x}_{\text{prompt}}$

a real joy. atmosphere alot dialogue Clone totally [MASK].



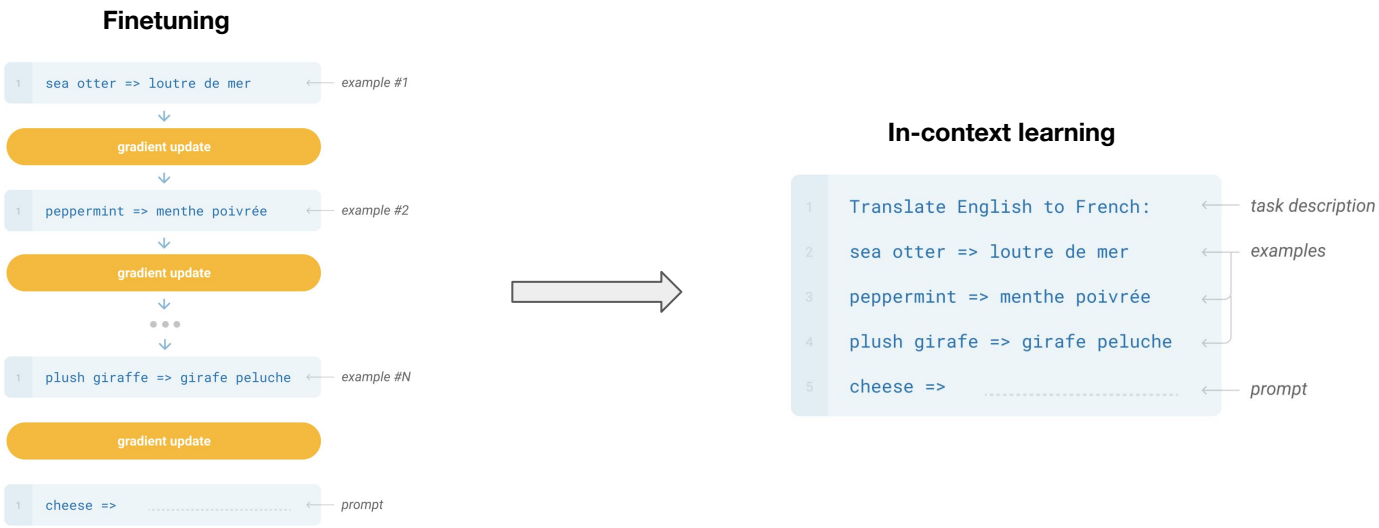
Shin et al; 2020

How to construct the prompts?

- Many other methods exist for searching for the best prompts
  - Both “natural” and “unnatural” prompts
- Most of these do not perform much better than manual prompt engineering
- But, choice of prompt is very important, especially for medium-scale models
- Soon: models which are trained to do better at prompts (instruction tuning / RLHF)

GPT-3: in-context few-shot learning

- GPT-3: decoder-only model (**175B** parameters) trained on over 600GB data
- **In-context learning (ICL)**: no gradient updates for learning the new task
- This is a special case of prompting – often improves accuracy



Brown et al; 2020

GPT-3: in-context few-shot learning

	SuperGLUE Average	BoolQ Accuracy	CB Accuracy	CB F1	COPA Accuracy	RTE Accuracy
Fine-tuned SOTA	**89.0**	**91.0**	**96.9**	**93.9**	**94.8**	**92.5**
Fine-tuned BERT-Large	69.0	77.4	83.6	75.7	70.6	71.7
GPT-3 Few-Shot	71.8	76.4	75.6	52.0	92.0	69.0

	WiC Accuracy	WSC Accuracy	MultiRC Accuracy	MultiRC F1a	ReCoRD Accuracy	ReCoRD F1
Fine-tuned SOTA	**76.1**	**93.8**	**62.3**	**88.2**	**92.5**	**93.3**
Fine-tuned BERT-Large	69.6	64.6	24.1	70.0	71.3	72.0
GPT-3 Few-Shot	49.4	80.1	30.5	75.4	90.2	91.1

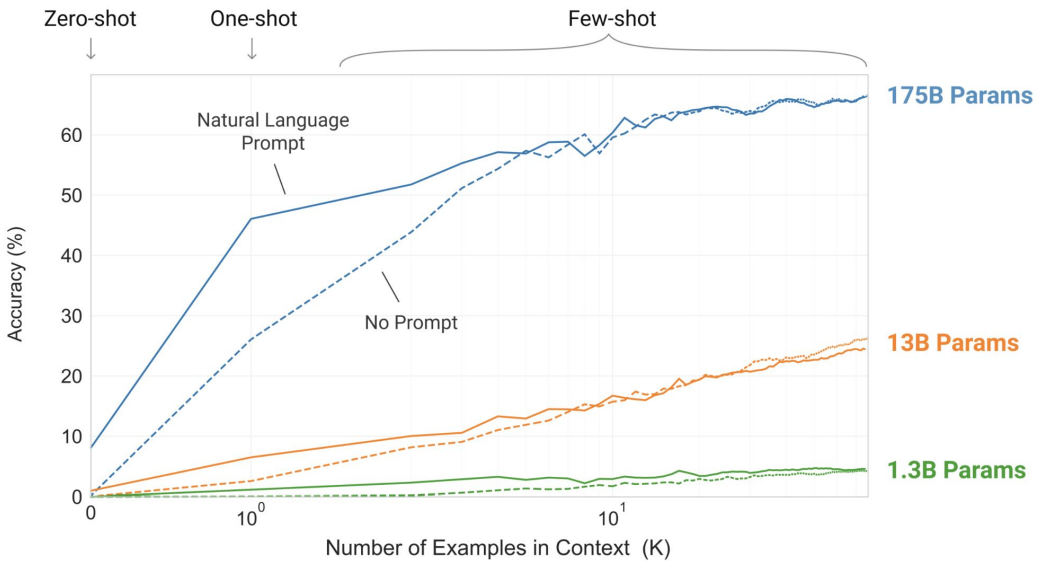
Table 3.8: Performance of GPT-3 on SuperGLUE compared to fine-tuned baselines and SOTA. All results are reported on the test set. GPT-3 few-shot is given a total of 32 examples within the context of each task and performs no gradient updates.

Brown et al; 2020

In-context learning emerges with scale

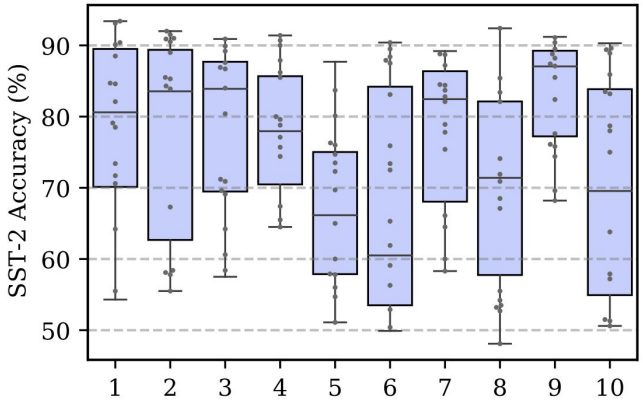
Task: remove random symbols from a word

s.u!c/c!e.s s i/o/n => succession



Brown et al; 2020

Choice of prompt matters



Different examples and different orders lead to very different accuracies

Calibrate before use Zhao et al, 2021

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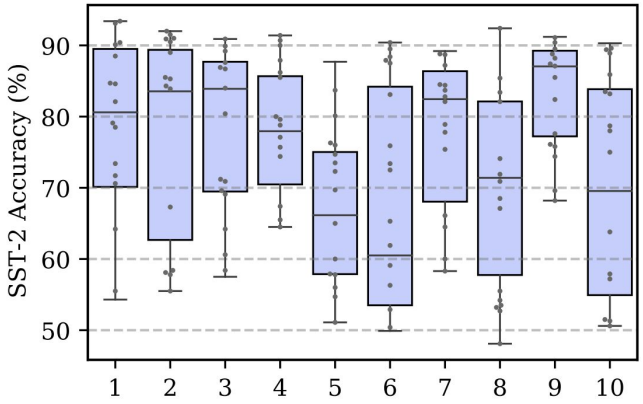

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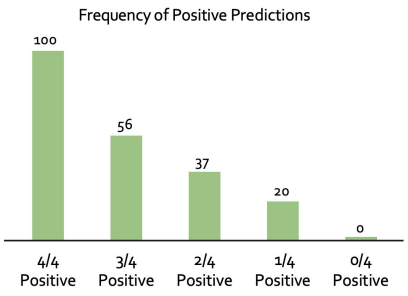

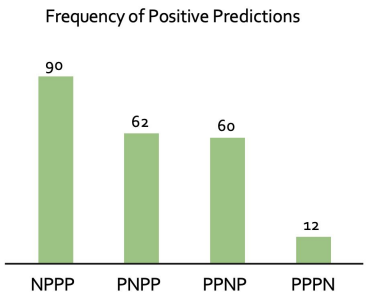

Model is biased towards majority labels and most recent labels

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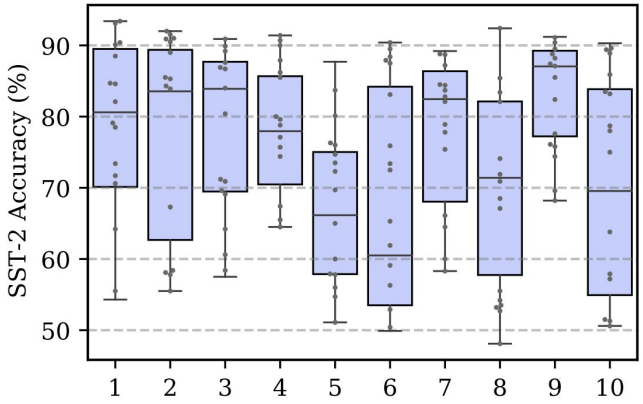

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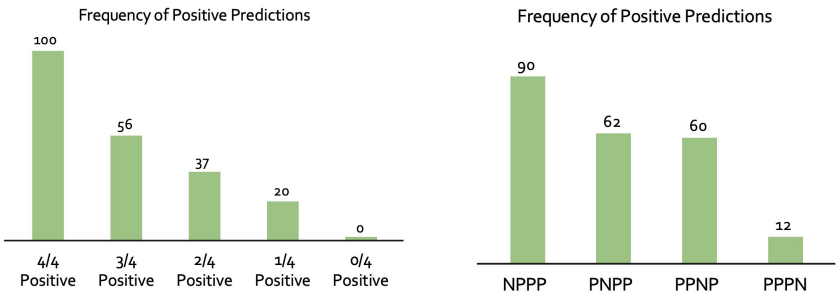


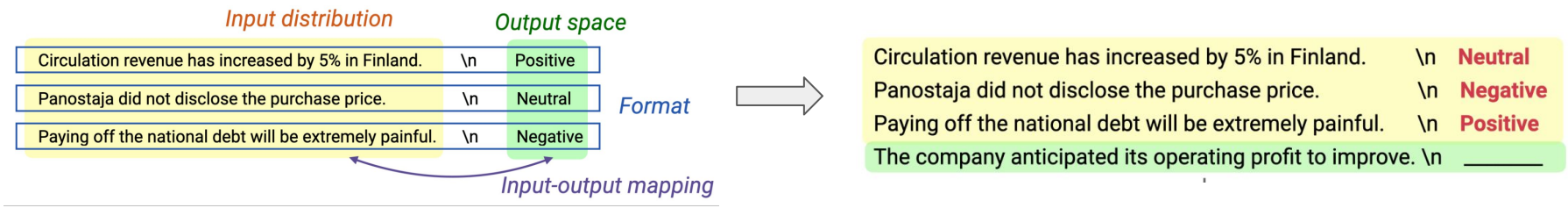
Model is biased towards majority labels and most recent labels

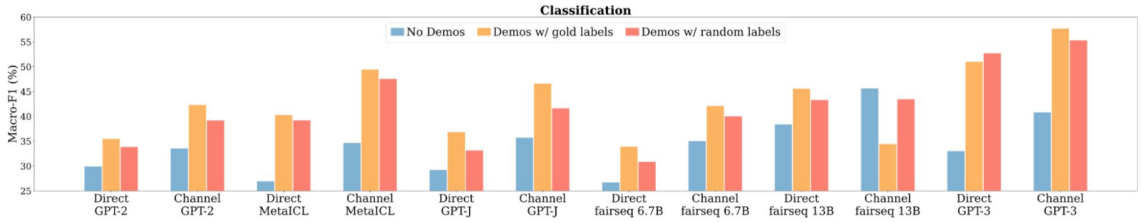
Note: these choices matter less as models are scaled up, and also if they are post-trained (next lecture)

Calibrate before use Zhao et al, 2021

A weirdness of ICL (Min et al; 2022)

- Flipping labels of the in-context examples has limited impact!





Credit: http://ai.stanford.edu/blog/understanding-incontext/

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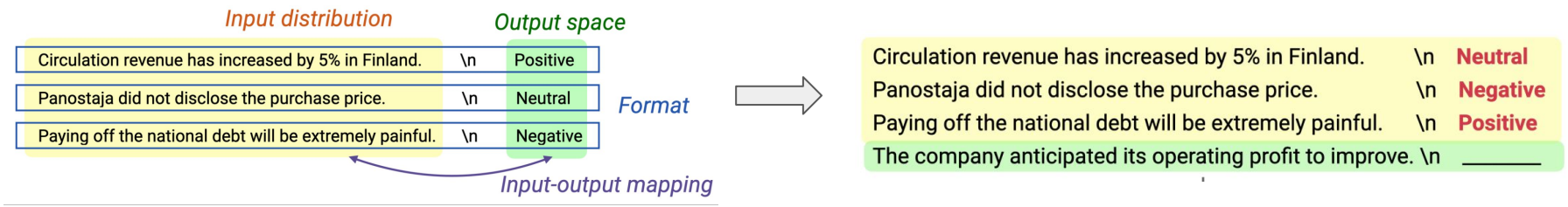

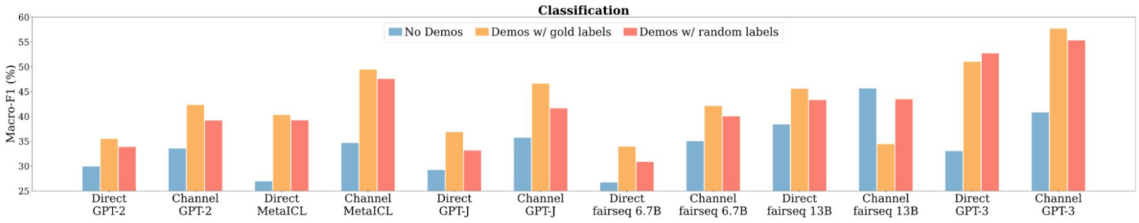

- This suggests that ICL is not learning in the traditional sense
  - Model is primarily recalling the knowledge already learned during pretraining

Credit: http://ai.stanford.edu/blog/understanding-incontext/

More difficult tasks?

Few-shot prompting is not enough for tasks which involve complex, multi-hop reasoning

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ✗

Wei et al; 2022

Chain-of-thought prompting

Add explanations which go through the step-by-step process of solving the problem

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Model Output

A: The answer is 27. ✗

Standard prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9. ✓

Chain-of-thought prompting

Wei et al; 2022

Chain-of-thought prompting

Add explanations which go through the step-by-step process of solving the problem

Date Understanding

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.

Sports Understanding

Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship."

A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer. So the answer is no.

Last Letter Concatenation

Q: Take the last letters of the words in “Lady Gaga” and concatenate them.

A: The last letter of “Lady” is “y”. The last letter of “Gaga” is “a”. Concatenating them is “ya”. So the answer is ya.

Coin Flip (state tracking)

Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up?

A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up. So the answer is no.

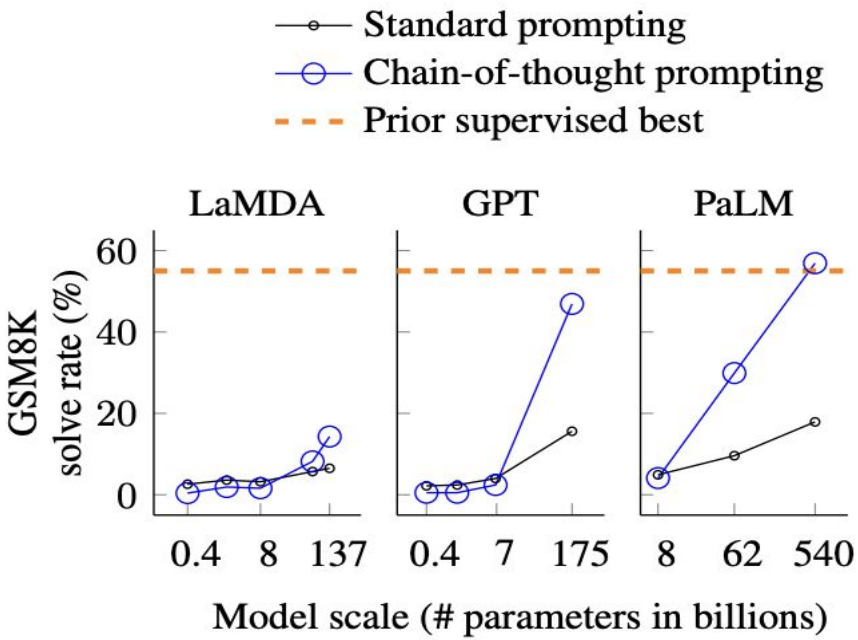
These are also known as “reasoning” or “thinking” traces

Wei et al; 2022

Chain-of-thought prompting

Chain-of-thought is also an emergent property of model scale

Middle-school math problems



Wei et al; 2022

Zero-shot chain-of-thought

We can also prompt the model to do chain-of-thought reasoning **without** providing labeled examples

Few-shot chain-of-thought

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are 16 / 2 = 8 golf balls. Half of the golf balls are blue. So there are 8 / 2 = 4 blue golf balls. The answer is 4. ✓

Zero-shot chain-of-thought

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let's think step by step.***

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

Kojima et al; 2023

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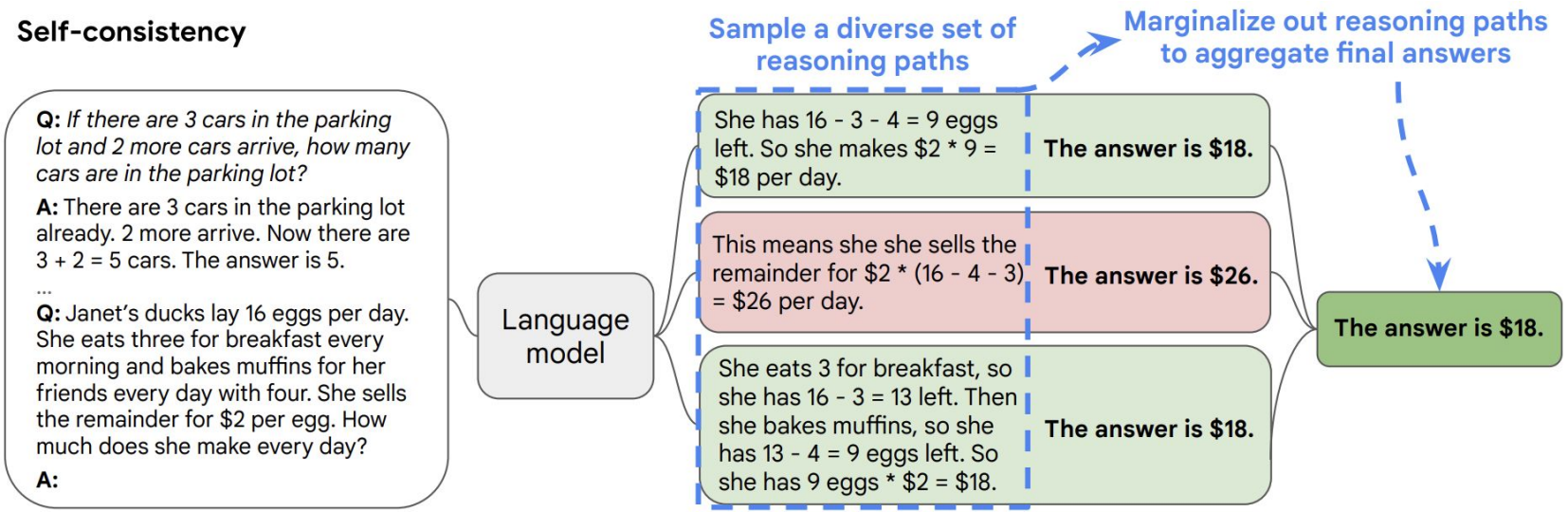
- Better than zero-shot or few-shot without chain-of-thought
- But few-shot chain-of-thought is still better

Kojima et al; 2023

Self-consistency

Instead of a single reasoning explanation, sample several and marginalize out to get the final answer



Wang et al; 2023

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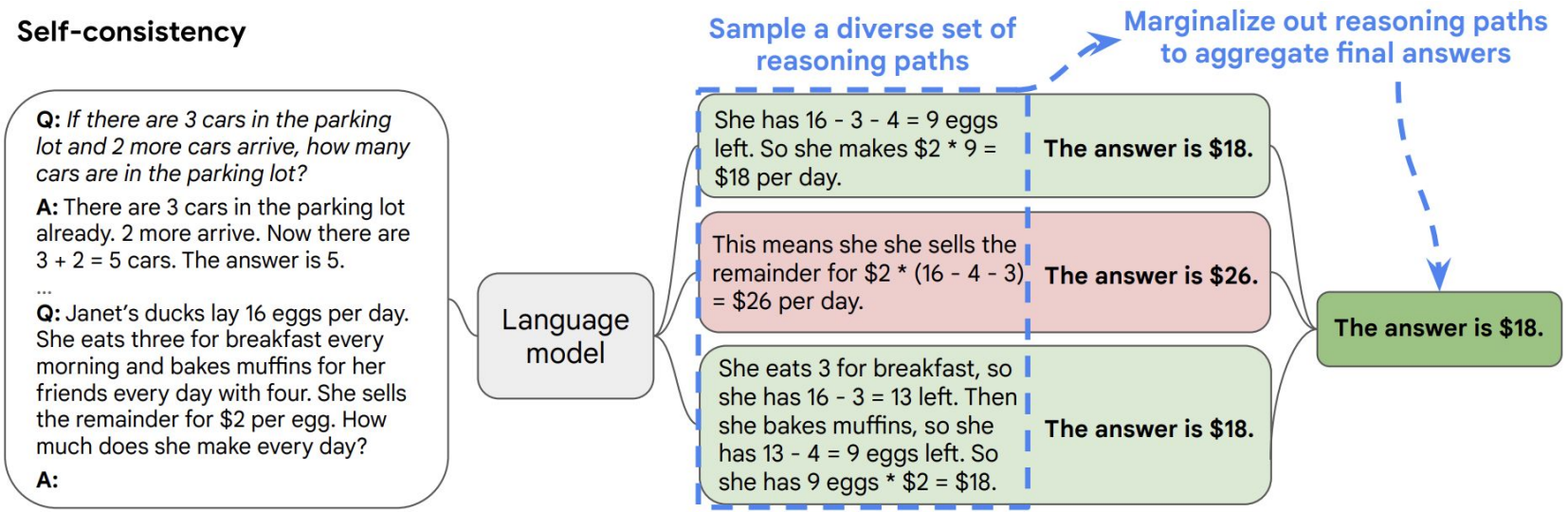

Improves GSM8K accuracy from 56.5 (chain-of-thought) to 74.4

Wang et al; 2023

Summary

- Prompting: formatting NLP tasks so that LMs can predict the output
- In-context learning: when prompts contain demonstrations (few-shot)
- Chain-of-thought: prompts instructing the model to reason or think before producing a final answer
- Self-consistency: sampling multiple outputs and taking a majority vote

That’s it!

Next lecture (Prof Monica Agrawal): Post-training